# Investments Homework Assignement (Course: Investments PSBV107NAMB)

**DEADLINE: 2023.12.22. 23:59 moodle upload**

Students can work alone or in group of two people.
Maximum 10 points can be achieved by submitting this homework.
Please upload a maximum 5 pages long pdf file to the moodle homework slot.

### Dynamic Hedging – a discrete approach

1. Choose any single stock that is a member of the S&P500 Index.

2. Choose any OTM (out-of-the-money) strike for a call option on the stock, expiry 15th December 2023. **(1 point)**

3. Number of **contracts = 10 + (KM1+KM2)/100** (please round it to the next integer number)

   KM1 = the total number of kilometers the group member #1 was born away from Budapest.
   KM2 = the total number of kilometers the group member #2 was born away from Budapest. (if you solve this homework alone, KM2 = 0)

   Please note that each contract is for 100 pieces of stocks, so the total notional of the option position will be 100 x contracts.

4. Assume you have sold the call option in the above mentioned number of contracts at the end of 10th November 2023. What was the option price? What is total amount of option premium (the total USD) you have collected? **(1 point)**

5. To hedge the delta risk of the short call position you buy some stocks at the closing price of 10th November 2023. How many stocks should you buy? **(2 points)**

6. Until expiry at the end of each day you calculate your total delta (delta of the short call + delta of the stocks) and you buy or sell some stocks in an amount that would make your total delta close to zero. Show the list of the required stock trades (stock amount, buy/sell, stock price, date).
   **(4 points)**

7. At expiry if the option is ITM you should assume that it is assigned. If it is ATM or OTM you should assume it is not assigned. After expiry if needed you should make a final stock trade to make your delta down to zero. How many stocks have you traded alltogether during the lifetime of your option position? What is the ratio of total stock traded compared to the notional of the option? **(1 point)**

8. After expiry you will have zero option position and you sould have zero stock position since you will hedge back all your delta risk to zero. However you probably will have some profit or loss after executing this strategy. What is the final result of your strategy? Why it is not zero? **(1 point)**